pfi

Python for Fixed Income

COURSE 2 · BOND MATH

Case Study: Convexity and the September Book

WEEK 2 · DAY 5

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Week 2 ends: one measure, one real job

Hands-on, 10.1 to 10.4: convexity

The four bonds of the week, settled 30 November 2026. Where duration's straight line misses, the convexity function, a check with three Excel PRICE cells, and a full reprice.

Case study, 10.5 to 10.10: the September book

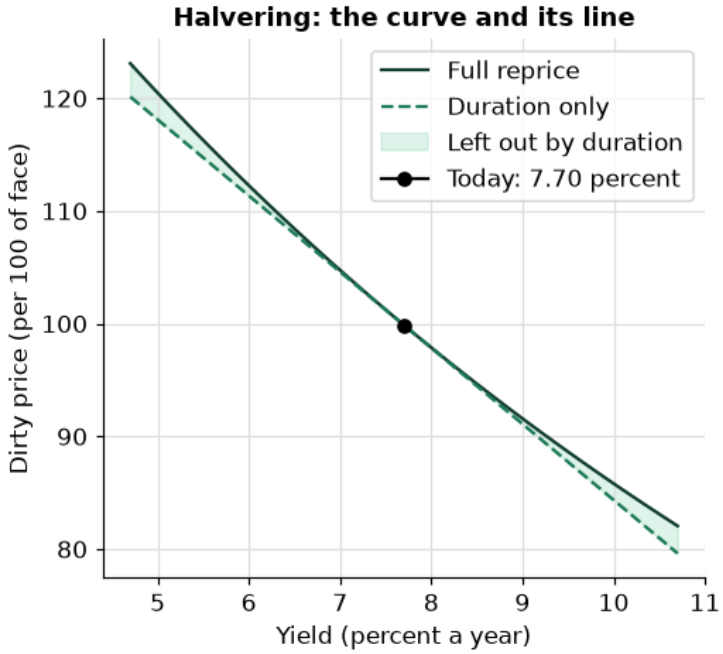
holdings.csv, 200 bonds. Price, accrued, yield, and duration rebuilt for every bond and checked with assert. Then the book's DV01 in dollars.

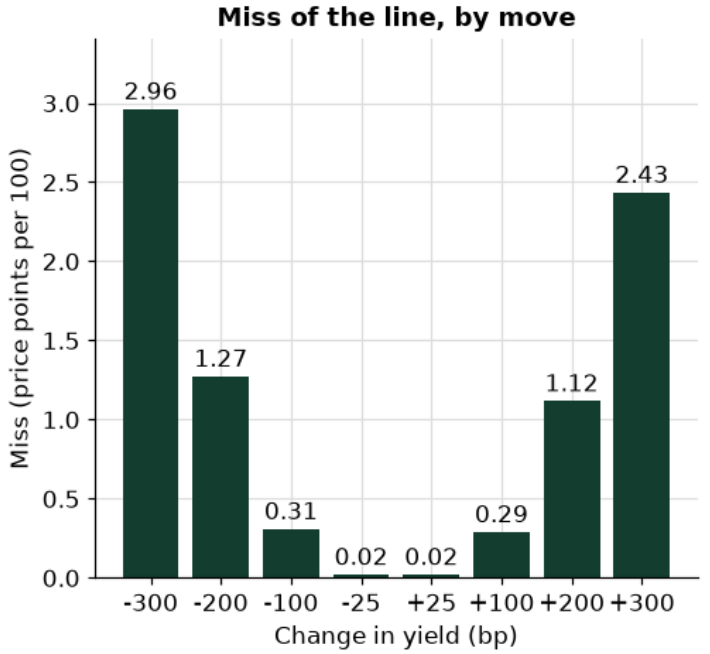
Week 2: dates, accrued, and risk

Day 6	Dates and coupon schedules
Day 7	Day counts and accrued interest
Day 8	Clean and dirty price
Day 9	Duration and DV01
Day 10	**Convexity and the September book**

Invented issuers and numbers. The day describes the book and never says a bond is cheap or rich.

Where duration's straight line misses





The miss is positive both ways, and three times the move is about nine times the miss.

Duration is a straight line laid against a curve. It works for small moves, not large ones.

Convexity: the bend, in years squared

```
# dP / P, with dy as a decimal
-modified_duration * dy + 0.5 * convexity * dy ** 2
```

Each cash flow adds CF x t x (t + 1/f) / (1 + y/f) ^ (f t + 2), and the sum is divided by the dirty price. Roughly time squared: long bonds have far more convexity.

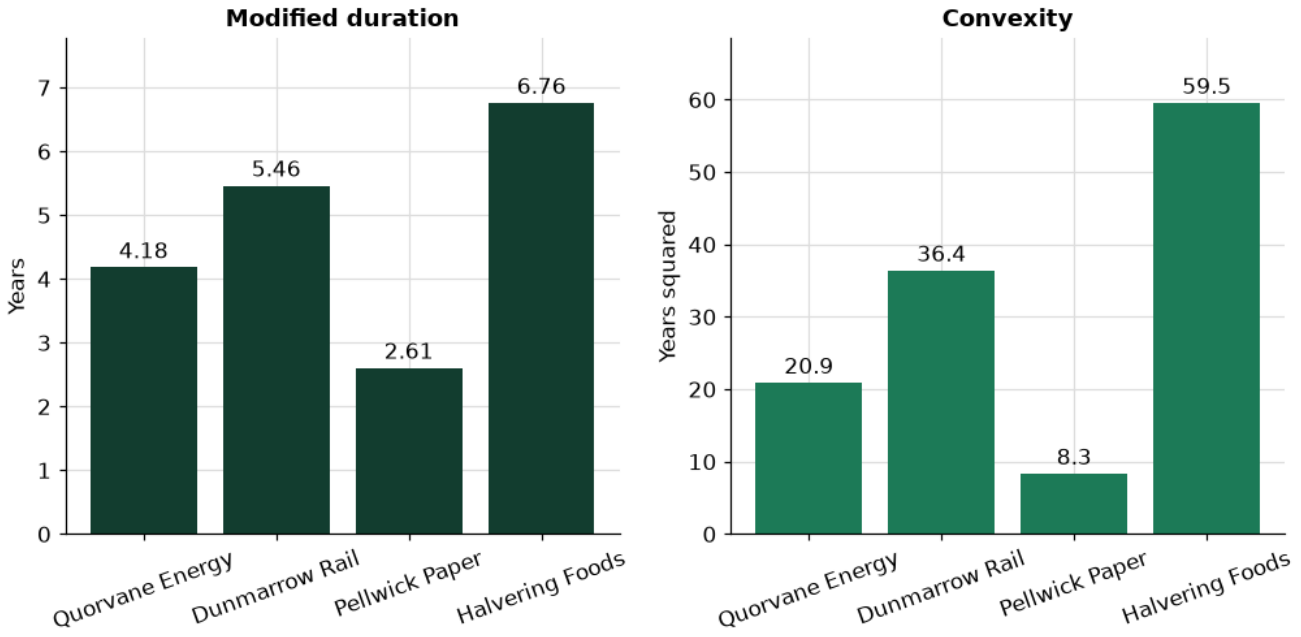
Other sources scale it differently: divided by 100, or halved. The course's number is unscaled. Halvering: 59.5474.

Bond	Mod. duration	Convexity
Pellwick	2.6051	8.3305
Quorvane	4.1804	20.8885
Dunmarrow	5.4623	36.3887
Halvering	6.7598	59.5474

Halvering against Pellwick: duration 2.6 times larger, convexity 7.1 times.

Units: years squared. A bullet bond's convexity is always positive.

Duration and convexity, drawn



Duration rises with the length of the bond. Convexity rises much faster: the longest bond bends most.

Excel has no convexity function: three PRICE cells

Price the bond at today's yield, 1 bp lower, and 1 bp higher. The curvature, over the dirty price, is a central difference. Accrued cancels out.

Excel side by side (tested in Excel)

A1	`=PRICE(DATE(2026,11,30),DATE(2036,9,30),7.5/100,7.7/100,100,2,0)`	98.622491
A2	`=PRICE(...,7.7/100-0.0001,100,2,0)`	98.690032
A3	`=PRICE(...,7.7/100+0.0001,100,2,0)`	98.555009
A4	`=100*7.5/100/2*COUPDAYBS(...)/COUPDAYS(...)`	1.250000
A5	`=(A2+A3-2*A1)/((A1+A4)*0.0001^2)`	59.547369

59.5474

Halvering, bondmath.convexity

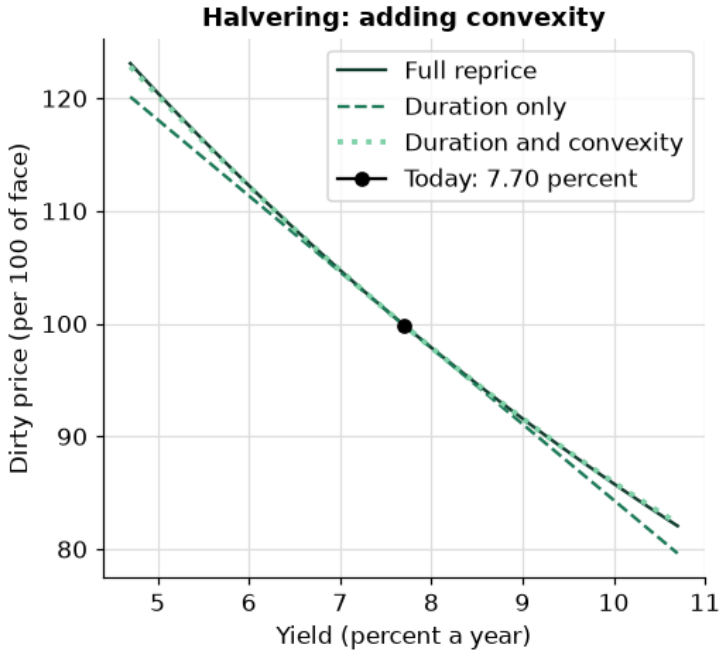
4.8e-6

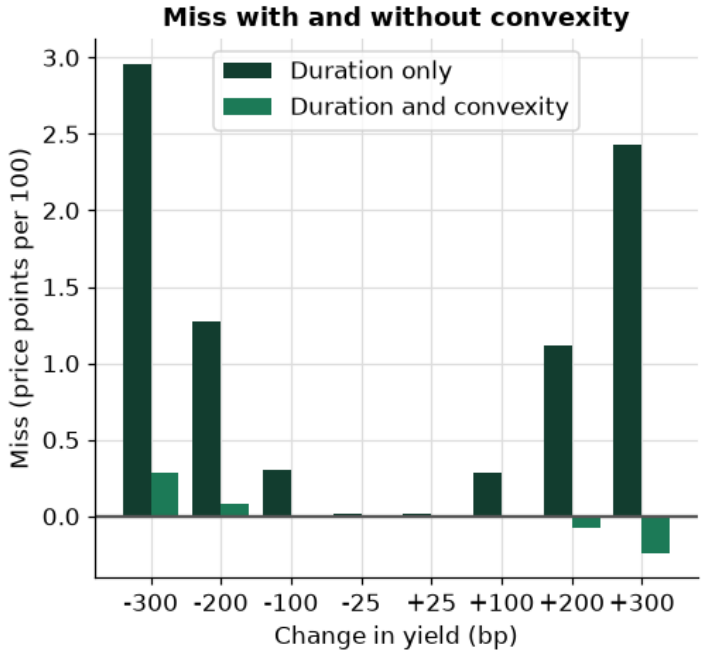
gap to Excel's central difference

28 passed

with the new test: every reference row, 1e-3

Duration plus convexity against a full reprice





With convexity the miss is small and changes sign. The full reprice is the answer.

+100 bp, Halvering: estimate -6.4620 percent, full reprice -6.4715.

The September book: check it, then add it up

Context

holdings.csv: 200 invented bullets as of 30 September 2026, semiannual, 30/360. Its price, accrued, yield, and duration are rounded to 3 decimals.

Objective, with your own module

1. price from yield_pct
2. accrued from the dates
3. yield_pct from price
4. duration, and a new convexity column
5. the benchmark, and what does not match
6. book DV01 in dollars, and its duration

Tolerances, with reasons

Column	Tolerance
price, accrued, duration	0.0005 + 1e-9
yield from rounded price	0.001 percent

Rounded to 3 decimals: off by up to 0.0005. A yield solved from a rounded price inherits it.

Two Predict questions. Every check is an assert: one bond out, and the notebook stops.

Price and accrued, every bond

```
# one bond at a time, with zip
book["price_calc"] = [bondmath.price(s, m, c, y)
    for s, m, c, y in zip(...)]
assert price_gap.max() <= ROUNDED_3
```

0.0004985

largest price gap, per 100

0.0005

largest accrued gap, per 100

All 200 bonds pass both checks.

Why the extra 1e-9

Bond 99113JAA2: the module's accrued is 0.0875, a rounding tie. The file holds 0.087. The true gap is exactly 0.0005.

```
0.0005000000000000004
```

Floating point stores neither number exactly. A tolerance of exactly 0.0005 would fail a bond that is right.

Yield from the file's price

Now the other way round: solve each yield from the file's rounded price with yield_to_maturity, and compare with yield_pct.

```
book["yield_calc"] = [
    bondmath.yield_to_maturity(s, m, c, p)
    for s, m, c, p in zip(...)]
```

Every price carries a rounding error of up to 0.0005 per 100. Where does it hurt the yield most?

Predict 1

Solved back into a yield, which bonds miss the file's yield by the most?

A) The longest bonds: the most cash flows

B) No pattern: rounding errors are random

C) The shortest: small duration, large yield error

Short bonds miss most; two bonds disagree

Predict 1: C, the shortest bonds

Bond	Mod. duration	Yield gap, %
99006GAC4	0.9270	0.000440
99026AAD1	1.0050	0.000433
99003DAG5	1.0170	0.000424

Average gap: 36 bonds under 3 years, 0.000139 percent; 79 bonds over 7 years, 0.000027 percent.

Duration: largest gap 0.0004982, all 200 pass. Then a convexity column, checked against Excel's central difference.

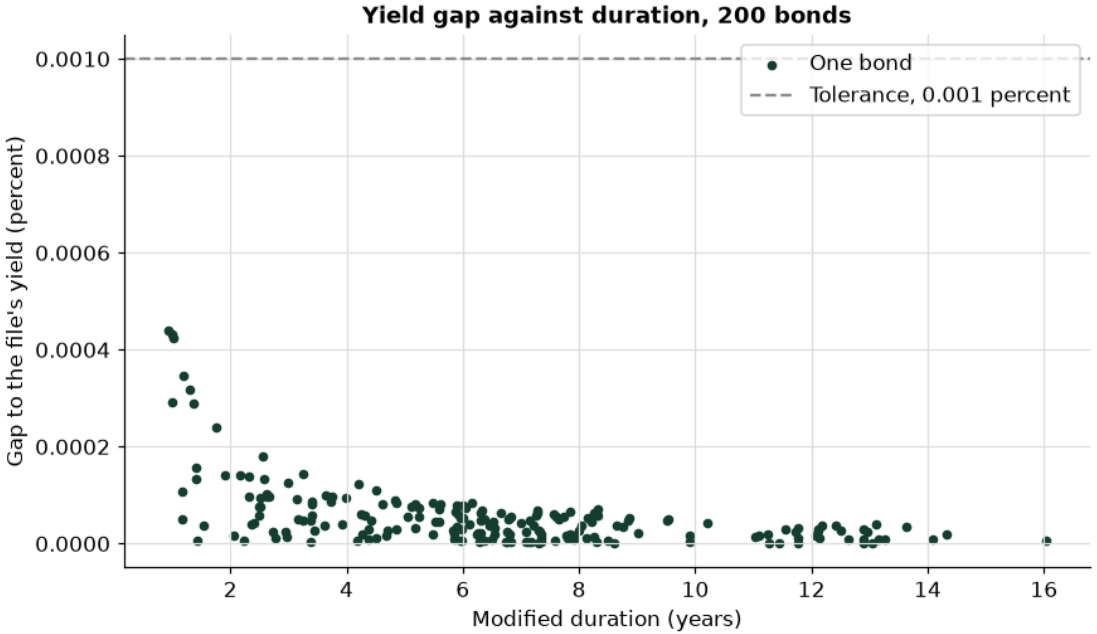
The benchmark: 821 bonds, two outside

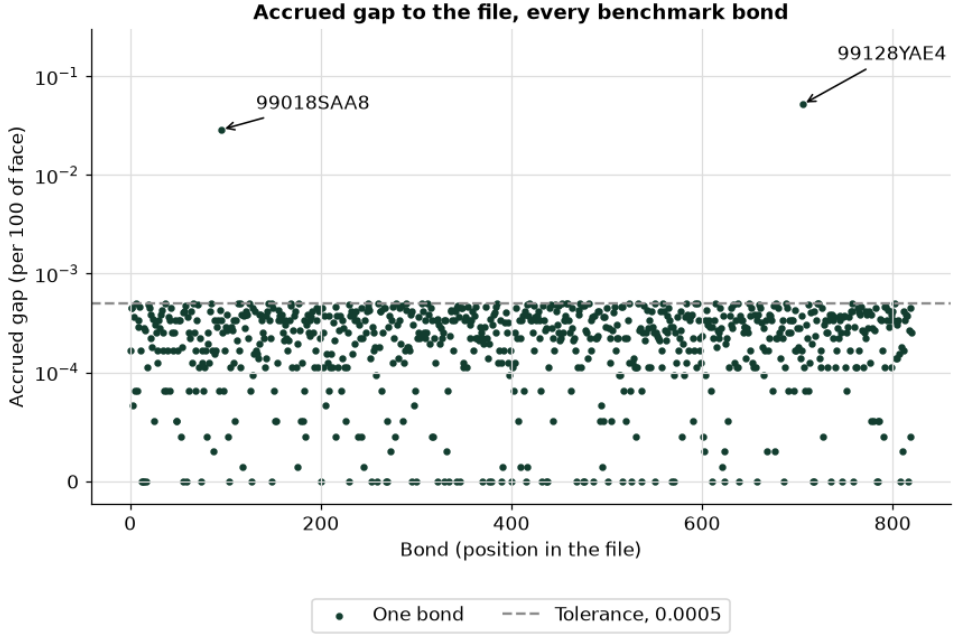
CUSIP	Accrued gap	Duration gap
99018SAA8	0.0289	0.0052
99128YAE4	0.0523	0.0056

Both mature on 28 February of a non-leap year. Excel's month-end rule puts coupons on 31 August; the data generator kept the 28th.

Both are conventions. Left out by name; a new failure still stops.

The yield gaps and the benchmark, drawn





Short bonds sit highest: a small price error is a large yield error when duration is small. All are below the tolerance.

Two bonds stand far above the tolerance: the 28 February maturities. Left out by name.

Book DV01 in dollars, and the book's duration

```
# dollars per bp, each position
dv01(...) * par_held / 100
# weights: market value, dirty price
(weights * duration_calc).sum()
```

306,996.11

book DV01, dollars per bp

6.2548

weighted duration, years

Face 489,000,000 dollars, market value 490,819,215.00 dollars, equal to the file's total to the cent.

Predict 2

Book duration x market value x 0.0001. How does it compare with the book DV01?

A) About 6 times larger: duration is in years

B) The same, up to the rounding of the prices

C) Much smaller: the weights add up to 1

One number in two units

306,996.21	306,996.11	-0.10
duration x MV x 0.0001	book DV01	difference, dollars

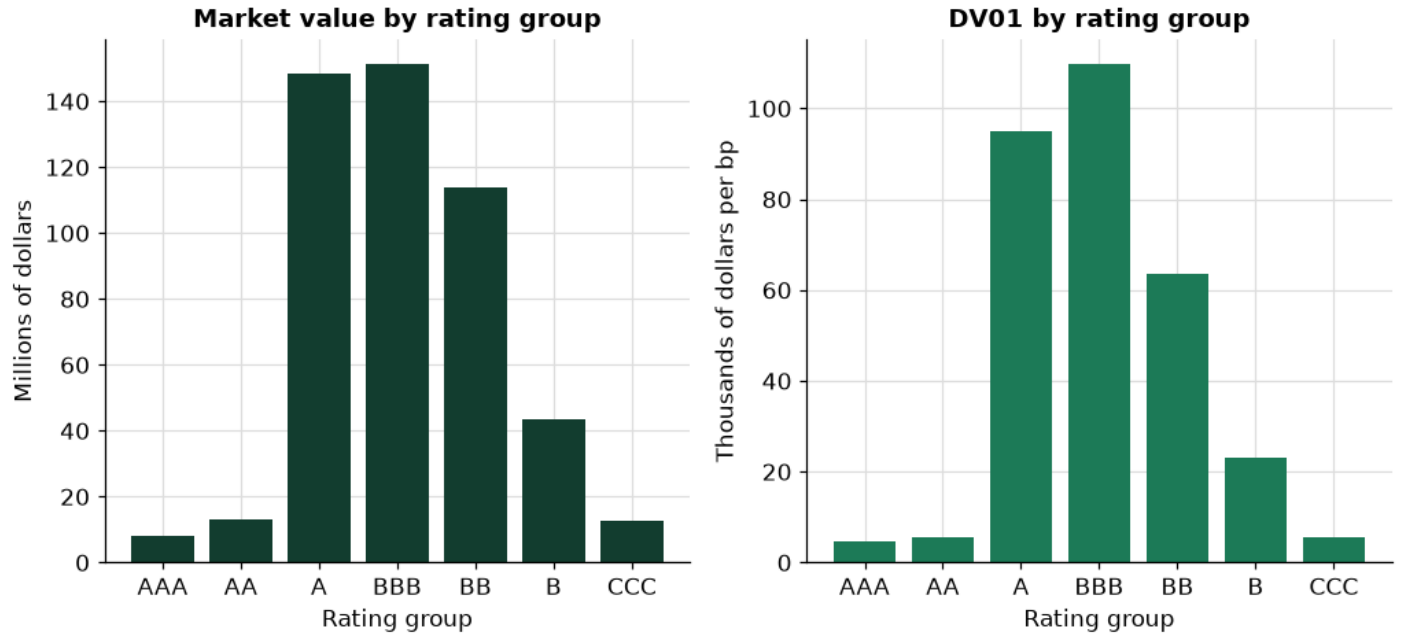
Each position's DV01 is duration x market value x 0.0001. Summed, the market values become weights. Years, or dollars per bp: one number.

Observations

- Every holdings bond reproduces at the course's tolerances; the largest gaps are rounding.
- Yields solved from rounded prices miss most on the shortest bonds.
- In the benchmark, only the two 28 February maturities differ: two conventions.

These describe the file and the book. None says which bond to buy or sell.

The book by rating group, drawn



The shapes are alike because DV01 is market value times duration times 0.0001. A description of the book, not of any group's value.

Today's Excel, every formula tested

Job	Excel (tested)	Returns
Convexity	`=(A2+A3-2*A1)/((A1+A4)*0.0001^2)`	59.547369
+100 bp estimate	`=(-MDURATION(...)*0.01+0.5*A5*0.01^2)*100`	-6.462029
+100 bp reprice	`=(PRICE(...,7.7/100+0.01,...)-A1)/(A1+A4)*100`	-6.471508
Book DV01	`=SUMPRODUCT(E2:E201,H2:H201)/100`	306,996.11
Market value	`=SUM(I2:I201)`	490,819,215.00
Weighted duration	`=SUMPRODUCT(I2:I201,J2:J201)/SUM(I2:I201)`	6.254772

The book sheet: one bond a row; H holds each DV01 per 100, I market value, J MDURATION. SUMPRODUCT multiplies two columns row by row and adds.

Excel's totals equal the module's. A1 to A5 are in section 10.3, A6 and A8 in 10.4.

End of week 2: tag it, compare it

Step	Command	What you see
1	git status --short, git diff --stat	M bondmath.py, M test_bondmath.py
2	git add, git commit	convexity and its test, committed
3	git tag week2, git tag	week1, week2
4	git log --oneline --decorate	tag: week1 and tag: week2
5	git diff --stat week1 week2	304 and 155 lines added

Two tags name two commits. diff between them is everything week 2 added.

The notebook's work folder is new, so the setup rebuilds the week1 commit: each file cut just before add_months.

Bring it home: paste, test, commit, tag

1

Your folder

Set-Location to my_bondmath, in your projects folder.

2

Paste

convexity at the end of bondmath.py, its test at the end of test_bondmath.py.

3

Test

python -m pytest -q: 28 passed, or more with your exercises.

4

Commit

git diff --stat, git add, git commit with a subject and a body.

5

Tag

git tag week2, git log --oneline --decorate, git diff --stat week1 week2.

Your folder has had its week1 tag since day 5. Copy fallen behind? Take the start-of-day files from today's setup cell.

In Colab: download bondmath.py and test_bondmath.py from the work folder under /tmp.

The AI check

The docstring is the prompt. The code that comes back runs, and has one convention error.

1

Docstring

Units, convention, and the Excel formula: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert against Excel's reference rows. Run it: it fails.

→

4

Fix

Find the convention it broke, fix it, run the test again.

Today: convexity left in periods squared, never divided by the frequency squared: four times too large.

Code that runs is not the same as code that is right.

What you can do now

- Say where duration's straight line misses, and why
- Compute convexity, in years squared, unscaled
- Check it with three PRICE cells, and estimate with both terms
- Reproduce a book's columns with assert, and add up its DV01
- Tag the end of a week, and compare two tags

Week 3, Day 1 Treasury curve interpolation: a yield split into curve and spread